

Key to more renewables

The rise of renewable energy sources like solar and wind brings a greater need for energy storage solutions due to several factors:

- **Intermittency.** The variable nature of renewable energy sources leads to intermittent power generation, creating a demand for energy storage to maintain a stable and reliable power supply.
- **Grid inertia.** Decommissioning fossil fuel plants reduces grid inertia, which is crucial for grid stability. Lithium-ion batteries, with appropriate power components, can potentially provide the necessary inertia.
- **Frequency and voltage stability.** Renewable energy integration can challenge grid frequency and voltage stability. Energy storage solutions are essential for managing these fluctuations and maintaining grid stability.

The increase in renewable energy sources introduces challenges such as intermittency, reduced grid inertia, and frequency and voltage instability, all of which drive the need for advanced energy storage solutions to ensure a stable and reliable power grid.

age, such as disaster resilience and off-grid power systems.

Pumped hydro. Pumped hydro storage remains a vital component of the energy storage landscape, providing large-scale, long-duration storage capabilities that are crucial for grid stability. While its deployment is limited by geographical and environmental considerations, ongoing research into innovative designs and site selection methodologies aims to expand the feasibility and reduce the environmental impact of new pumped hydro projects.

Measuring energy storage project costs

The levelised cost of storage (LCOS) is a crucial metric for assessing the economic viability of energy storage projects. It encompasses all costs associated with a project, including capital expenditures (capex), operation and maintenance (O&M) costs, and replacement costs, divided by the total energy output. This calculation provides a standardised way to compare different storage technologies by presenting costs in terms of US dollars per kilowatt-hour (\$/kWh).

When evaluating the LCOS, it is essential to consider the specific characteristics of each storage technology. For instance, lithium-ion

batteries require accounting for their cycle life and replacement frequency, whereas compressed air energy storage systems need to factor in the O&M costs for their mechanical components. By using the LCOS metric, developers can make informed decisions about the most cost-effective storage solution for their needs.

The upfront costs of energy storage projects can be categorised into several key components:

Storage system costs. For electrochemical systems, this includes the cost of cells or electrolytes, while for mechanical systems, it involves the cost of reservoirs or caverns.

Power components. Expenses related to power electronics, HVAC systems for temperature control, and housing for batteries.

Construction and engineering costs. Costs associated with building the storage facility and its engineering design.

Grid connection costs. The cost of connecting the storage system to the electrical grid.

The cost efficiency of an energy storage system is influenced by various factors, including the technology type, operational costs, efficiency, and lifespan. For example, flow batteries have significant costs in electrolytes and power components, while compressed air systems have

high excavation and construction costs.

O&M costs and replacement costs play a crucial role in determining the total cost of ownership of an energy storage system. For example, while lithium-ion batteries may have lower upfront costs, their shorter cycle life compared to flow batteries can result in higher replacement costs over time. Therefore, it is vital to consider these costs when evaluating the economic viability of different energy storage technologies for specific applications.

Texas energy storage: Meeting peaks, creating opportunities

The need for significant peaking capacity in Texas, highlighted by the winter storm outages, has driven a surge in energy storage deployment. The state's experience with massive power outages due to frozen infrastructure and reduced solar generation has underscored the importance of having dispatchable assets for grid resiliency. As a result, lithium-ion batteries have been increasingly implemented in Texas to provide reliable power during peak demand periods and outages.

Local energy requirements also play a crucial role in shaping grid needs and the utilisation of energy storage. For example, a grid with a high percentage of distributed solar energy may require energy storage solutions at the customer's home or distributed storage to address regional issues. Conversely, a region reliant on offshore wind may need energy storage at grid connections to prevent transmission congestion and longer-duration storage to manage the seasonal intermittency of wind.

The growth in energy storage is primarily driven by the ancillary services market and the expanding microgrid sector. Ancillary services, although becoming saturated in some areas, still offer lucrative opportunities for short-duration energy storage.